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SURVEY

Plant Leaf Disease Detection, Classification, and Diagnosis Using Computer Vision and Artificial Intelligence: A Review

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ABSTRACT Agriculture is the ultimate imperative and primary source of origin to furnish domestic income for multifarious countries. The disease caused in plants due to various pathogens like viruses, fungi, and bacteria is liable for considerable monetary losses in the agriculture corporation across the world. The security of crops concerning quality and quantity is crucial to monitor disease in plants. Thus, recognition of plant disease is essential. The plant disease syndrome is noticeable in distinct parts of plants. Nonetheless, commonly the infection is detected in distinct leaves of plants. Computer vision, deep learning, few-shot learning, and soft computing techniques are utilized by various investigators to automatically identify the disease in plants via leaf images. These techniques also benefit farmers in achieving expeditious and appropriate actions to avoid a reduction in the quality and quantity of crops. The application of these techniques in the recognition of disease can avert the disadvantage of origin by a factious selection of disease features, extraction of features, and boost the speed of technology and efficiency of research. Also, certain molecular techniques have been established to prevent and mitigate the pathogenic threat. Hence, this review helps the investigator to automatically detect disease in plants using machine learning, deep learning and few shot learning and provide certain diagnosis techniques to prevent disease. Moreover, some of the future works in the classification of disease are also discussed.

INDEX TERMS Deep learning, diagnosis, image processing, machine learning, plant disease.

I. INTRODUCTION

The agriculture and food organization of the United Nations disclosed that the total number of starved people across the world has been growing deliberately since 2015 [1]. The ongoing estimation reveals that around 680 million people are starved and they comprise under 9% of the population around the world. This indicates an expansion of 10 million in a year and approximately 120 million in 10 years. Around more than 85% of people confide in agriculture in the world [2]. Accordingly, the food industry and agriculture demand adequate farming mechanisms. Further, living creatures accept oxygen from plants in the balancing environment by photosynthesis. The plant disease that affects the leaves may lead to death or bad health of the plant and abort to implementation of plant food. Also, plant disease has a gigantic effect on developing food crops. In 1845 potato (Irish famine) resulted in 1.2 million deaths [3]. Some of the laboratory approaches used for plant disease are immunosorbent enzymes, isothermal amplification, and polymerase reaction. Hence, early

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detection, management, and prevention of disease in plants are precisely essential. Nonetheless, disease detection of plants in the enormous field is a very complicated task that involves trained manpower and optical examination of leaves [4]. Farmers observe the symptoms of diseases on plant leaves with the naked eye and diagnose plant diseases based on experience, which is laborious, time-consuming, and requires special skills [5]. The objective of a disease detection system is to support non-expert users, i.e. non-pathologist and non-botanists. Therefore, various automated techniques have been developed using image processing, machine learning, deep learning, few short learning are discussed in this review paper. The agriculture practical applications are not sufficient using conventional ML approaches due to robustness and laboratory conditions [6], [7]. The deep learning in recent times has built exceptional in the classification of images for plant disease [8], [9]. The approach required a large number of samples to meet the challenges in time. Also, all the images are fundamentally annotated by pathologists and ethnologists. Since these efforts are laborious and time-consuming, deep learning applications in disease detection are costly [10], [11]. FSL facilitates learning from machines using a limited number of labeled databases [12], [13]. The experiment's purpose and problem complexity determine the number of shots required. Various pathogens [14], [15] cause the disease a diagnosis by DNA (Deoxyribose nucleic acid), PCR (Polymerase chain reaction), MPG (Modified panchayat mixture), ELISA (Enzyme-linked immunosorbent assay), FISH (Fluorescence in situ hybridization) and IF (Immunofluorescence) method. Therefore, the objective of this review paper is to give a comparative analysis of machine learning, deep learning, few short learnings in plant disease detection and also review various segmentation, feature extraction, and classification techniques. Also, certain molecular diagnosis tools are reported.

II. PHYTOPATHOLOGY

The study of plant pathogens, responsible pathogens their disease, mechanism, control methods of disease and to diminish their brunt on plants is referred to as Phytopathology [16]. As a result, it is a system for dealing with a plant's life cycle. Phytopathology is a Greek letter word that means plant (Phyto), disease (Patho), and knowledge (Logo). The basic objectives of phytopathology are a study of the source, and causes of plant disease as biotic or abiotic (etiology), a mechanism study of development for disease (pathogenesis), the interaction between plant disease and pathogen (epidemiology), and development management for reduction of radiation and losses control management as shown in Fig. 1.

Phytopathology comprises of subdomain under agriculture science and consists of elemental learning of microbiology, physiology, nematology, virology, anatomy, bacteriology, mycology, genetic engineering, botany, meteorology, climatology, and molecular biology indicated in Fig. 2.

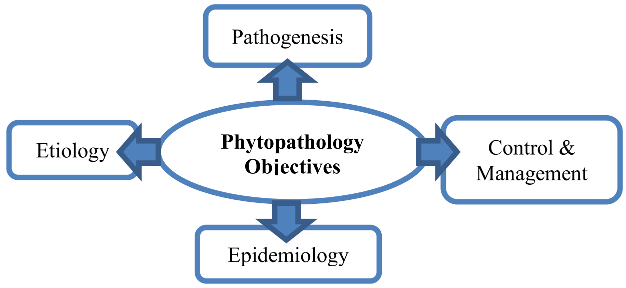

FIGURE 1. Phytopathology objectives.

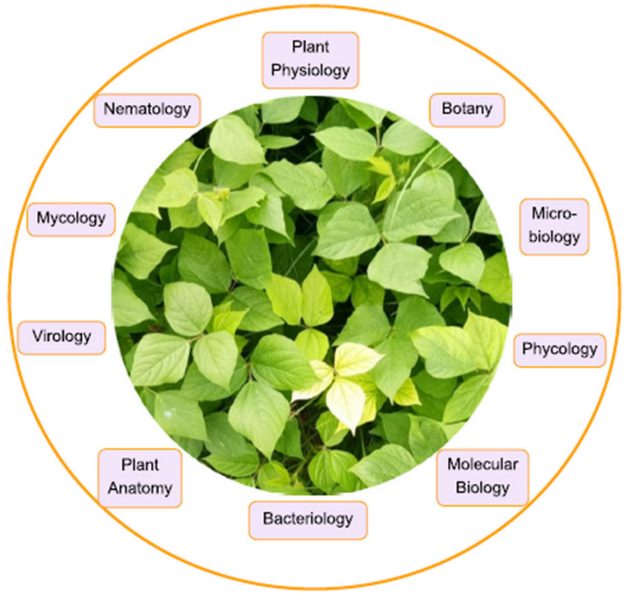

FIGURE 2. Subdomains of phytopathology [16].

III. PLANT DISEASE/TYPES AND SYMPTOMS

The behavior or physiology of the plants in terms of abnormality results in plant disease. It is due to biotic or abiotic agents as indicated in Fig. 3 [16]. Biotic disease is generated from infectious agents whereas abiotic disease is generated from non-infectious agents. The abiotic disease is usually avoidable due to its non-transmissible lesser hazardous nature. Thus, in this manuscript, biotic disease is examined.

- Bacterial Disease- Plant bacterial disease occurs due to water-soaked results in a little green blemish. These lesions augment and then emerge as dead dry blemishes (spots) represented in Fig. 4(a). Example: The foliage consists of water-soaked black blemish or brown leaf spot or halo yellow with equivalent size. The blemish appears as dappled beneath dry conditions. Generally, Bacterial wilt occurs in brinjal crops due to which the whole plant falls [17].
- Viral Disease- Plant viral disease is the most burdensome disease to investigate among all the plant diseases. The virus observed on the plant has no indication or it looks similar to herbicide bruise as well as nutrient

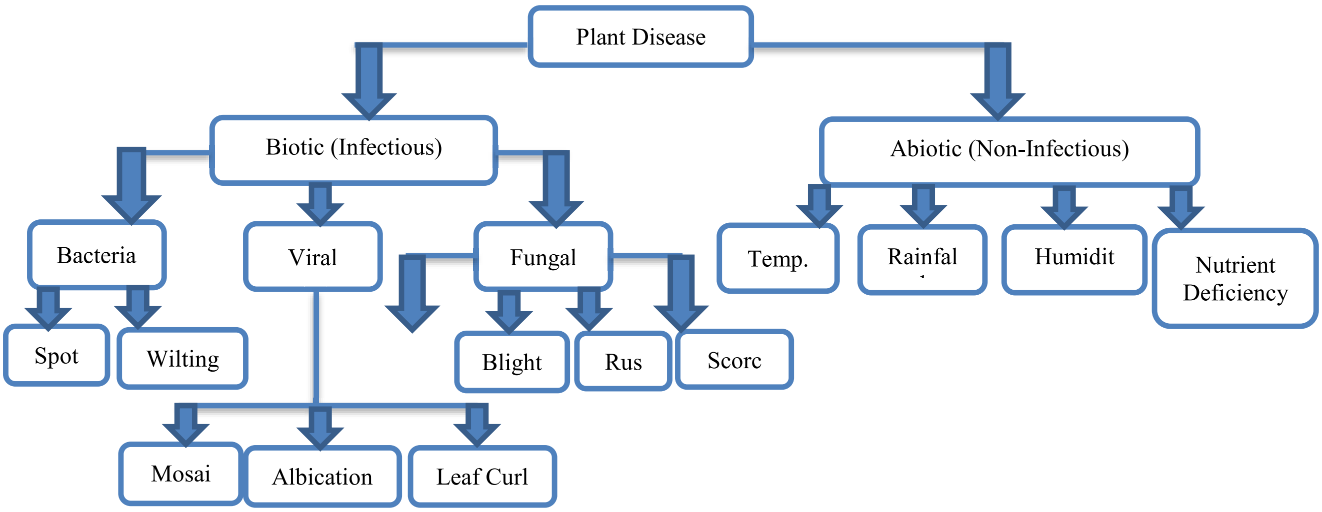

FIGURE 3. Classification of plant disease in distinct categories [16].

FIGURE 4. (a) Bacterial blemish [19] (b) Viral Mosaic [20] (c) Late Blight [22] (d) Early Blight [22] (e) Rust.

TABLE 1. Distinct disease in different plants.

Author's	Plant Name	Bacterial Disease	Viral Disease	Fungal Disease
Zhang et al. 2019 [19] Kianat et al. 2021 [20] Agarwal et al. 2021 [21]	Cucumber	Brown blemish, Angular Blemish, Target blemish	Mosaic, Yellow blemish	Black blemish, Gray mold
Shrivastava et al. 2019 [22] Chen et al. 2021 [23]	Rice	Streak, Blight	Black Dwarf Streaked	Smut False
Sun et al. 2021 [24]	Maize	Streak, Stalk	Crimson, Dwarf	Rust
Ferentinos 2018 [25] Abbas et al. 2021 [26]	Tomato	Canker	Curl leaf yellow	Late/ Early Blight

failure [17]. The most commonly found viral diseases are in beetles, leafhoppers, aphids, and whiteflies such as mosaic viral disease indicated in green or yellow stripes for foliage as shown in Fig. 4 (b).

- Fungal Disease- Plants fungal disease affects various components of plants i.e., sclerotium wilt, common crown rot, stem rust, eyespot (sheath or stem), rust, blight (leaves), ergot (spikes) and carnal bunt, black point (seeds). Phytophthora fungus caused by late blight emerges on former leaves such as gray-green blemish, waterlog shown in Fig. 4(c). This fungus is caused due to climate change in the form of wet and dry. As the late blight disease is cultivated, these blemishes get dark and fungus in white form grows on the surface [18]. Alternaria fungus caused by early blight, emerges on former leaves in the form of tinny brown blemish with bull's eye arrangement in the form of concentric rings as shown in Fig. 4(d). Rust fungus develops on ripe plant leaves on the curtailed surface as shown in Figure 4(e). This blemish becomes black after green-yellow.

TABLE 2. Distinct plants, their disease and responsible pathogen [16].

Plant	Disease	Pathogen	Symptoms
Apple	Scab	Pomi Spilocaea	Brown-Gray on leaf
	Rot	Malorum Sphaeropsis	Dark Brown on leaf
	Rust	Sporangium	Yellow pale on leaf
Cherry	Mildew	Clandestina	Gray powder on leaf
Corn	Gray Spot	Cercospora	Rectangle lesions
	Rust	Sorghi puccinia	Red pustules on leaf
	Light blight	Tutcica setosphaeria	Elliptical lesions
Grape	Rot	Bidwellii guignardia	Red borders on leaf
	Measles	Aleophilum	Necrotic stripping
	Isariopsis blight	Angulata brachypus	Coalesce lesions
Peach	Spot	Arboricola Xanthomonas	Clustered lesions
Potato	Early blight	Solani Alternaria	Brown lesion
	Late blight	Infestans phytophthora	Dark greeb spot
Tomato	Septoria spot	Lycopersici	Foliage
	Mosaic	Mosaic virus	Mottle green leaf
Orange	Green Citrus	Bacteria Motile	Precipitate Demolition
Strawberry	Scorch Fungus	Diplocarpon	Brown edges
Squash	Mildew	Xanthii podosphaers	White powder

The above-described and mentioned disease symptoms are distinct and very less compared to other conventional diseases in plants. That means the equivalent disease symptom may arise due to infectious or non-infectious diseases. Some of the distinct diseases in plants are indicated in Table 1.

Therefore, accurate diagnosis of plant disease is troublesome for the identification of pathogens using particular symptoms.

The artificial intelligence techniques, for detecting the performance of disease detection in plants essentially depend on the feature extraction as well as a classification for disease [27], [28], [29]. These extracted selected features reveal several disease symptoms. Table 2 represents a manifestation of several typical diseases in plants with specific symptoms. The following Table will help the investigator to identify the accurate features which would further lead to high classification accuracy.

IV. PLANT DISEASE DETECTION SYSTEM

The artificial intelligence techniques are addressed to enhance agriculture production by bestowing in plant disease monitoring. Various review research papers have been published [30], [31], some of them focus on particular approaches while others are centered on particular diseases. A detailed summarization of plant disease detection, classification, and diagnosis is not available. Therefore, this review focuses on distinct techniques used by distinct researchers that utilize machine learning (ML), deep learning (DL), few-shot learning (FSL), and soft computing techniques with image processing for RGB as well as hyperspectral images. Also, certain molecular techniques have been established to prevent and mitigate the pathogenic threat.

A. MACHINE LEARNING USING IMAGE PROCESSING

The categorization of disease detection in plants using machine learning with image processing is done in the following sequential steps: Image Acquisition, Image Preprocessing, Image Segmentation, Feature Extraction and selection, and Classification [32]. In this, firstly images of plants (roots, branches, stems & leaves) are captured. Next, image pre-processing i.e., noise removal or blur or any other distortion is removed using techniques like scaling, stretching, smoothening, transformation, and rotation. Next, segmentation to extract the infected region using techniques like thresholding, and clustering is done. Then this segmented region is used to extract features for classification. In this section, the requirement of all these steps is mentioned and the approaches proposed in the literature are also summarized.

1) IMAGE ACQUISITION

The initial gait for any machine learning system is image acquisition. This incorporates image capture or redeems images from repositories. The image quality highly leans on the camera and its acclimatization which leads to disease detection accuracy of the system [33], [34]. The captured image might dwell in the undesired background, adumbration, and noise. To improve the accuracy of the system or to extract the affected part firstly background removal and noise removal are required. Subsequently, other than RGB images from normal cameras, certain specific cameras are used to capture hyperspectral, thermal, and fluorescent images. Various researchers use distinct datasets for plant disease detection systems as indicated in Table 3. The presence of shadow, light, insufficient lighting, and intricated background is burdensome due to uncontrolled/real-time images as compared to laboratory surroundings conditions. The quality of the image captured robustly depends upon the techniques and equipment used. Consequently, the system performance is influenced by the image acquisition stage.

2) IMAGE PRE-PROCESSING

The crucial step in a machine learning system is image processing. It helps to improve the quality of the image that is degraded due to distortions, turbulence, or shadow [56]. Majorly, the dataset is captured in uncontrolled environments or real-time situations. Therefore, the preprocessing step is necessary before feature extraction enlarges the estimated accuracy of the system which results in reduced processing time. The process also reduces the processing time by applying operations such as crop, and resize. The literature utilizes image enhancement, cropping, and elimination of background as extensively used preprocessing techniques. These operations applicability depends on the image quality. Various researchers use unique techniques for preprocessing are indicated in Table 4. Image preprocessing along with augmentation is used to create more images of the data on the existing data for several deep learning techniques that require large amounts of image data. The processing augmentation techniques are noise injection, flipping of image, gamma correction, scaling, rotation, resize, cropping, random shift, zoom, and image transformation such as contrast enhancement and brightness.

3) IMAGE SEGMENTATION

The region of interest or undesired region can be segregated using image segmentation. This approach intends to separate the part having a deformity. This analysis of the representation of the image is simple and further meaningful to discriminate the affected and unaffected parts. It is the most important technique among machine learning techniques as critical issues and techniques in the analysis of an image. The several challenges faced such as erroneous boundaries of the affected region, shadow, illumination, and complex background [74]. The segmentation process is classified into two main divisions: first is conventional methods such as thresholding, region growing and edge detection other is computational techniques such as fuzzy logic, genetic algorithm, and neural network. Generally, computational techniques perform better than conventional techniques. The segmentation process is crucial for the extraction of features. Various researchers use unique techniques for segmentation are indicated in Table 5.

TABLE 3. Details of the dataset available used by various researchers.

	Dataset Name	Authors
Open accessible dataset	APS image dataset	Mohanty et al. 2016 [33]
	Plant Village Image dataset	Mohanty et al. 2016 [33]
	Computers and Optics in Food Inspection (Cofi) laboratory image dataset	Arnal Barbedo et al. 2019 [34]
	Digipathos images (PDDB)	Arnal Barbedo et al. 2019 [34]
	IRRI Dataset	Bashir et al. 2019 [35]
	INIBAP leaf Dataset	Camargo & Smith 2009 [36]
Self-created dataset	-	Karadag et al. 2018 [37]
	-	Coulibaly et al. 2019 [38]
	-	Pantazi et al. 2019 [39]
	-	Fuentes et al. 2017 [40]
	-	Shrivastava and Hooda 2014 [41]
	RoCoLe	J. Parraga Alava. et al. 2019 [42]
		K. Tian, et al. 2019 [43]
Multiple Crop dataset	Citrus dataset	Zhang and Meng 2011 [44]
	Grapefruit Grove	Pydipati et al. 2006 [45]
	-	Masazhar and Kamal 2018 [46]
University Agriculture	-	Deshapande et al. 2019 [47]
	-	Pujari et al. 2016 [48]
	-	Abed and Esmaeel 2018 [49]
Dataset using spectral devices	-	Azadbakht et al 2019 [50]
	-	Rothe and Kshirsagar 2015 [51]
Hyperspectral imaging	-	Abdulridha et al 2019 [52]
	-	Zhang et al 2018 [53]
Charge Couple device camera	-	Huang 2007 [54]
	-	Yao et al 2009 [55]

TABLE 4. Details of pre-processing technique used by various researchers.

Techniques Used		Author's
Color Space Conversion	Enhancement Filtering, Background reduction	Kaur et al. 2018 [56] Das et al. 2020 [57]
	RGB, HSV, HSI, YIQ, L*a*b, grayscale	Khot et al. 2016 [58], Dhingra et al. 2017 [59], Cruz et al. 2018 [60]
	L a* b*, HSV	Rothe and Kshirsagar 2015 [37], Vidyaraj and Priya 2016 [61], Kaur et al. 2018 [37], Pantazi et al. 2019 [39]
	YCbCr, CIE	Kai et al. 2011 [62], Chaudhary et al. 2012 [63], Joshi and Jadhav 2017 [64]
	H, I3a, I3b	Camargo and Smith 2009a [36]
	CIELuv	Ganeshan et al. 2017 [65]
	La*b*, Luv, YCbCr	Meunkaewjinda et al. 2008 [66]
Image Enhancement Techniques	Denoising using Mean and median filtering	Yao et al. 2009 [41], Rao & Kulkarni 2020 [67]
	Sharpening using Gaussian and Laplacian filtering	Asfarian et al. 2013 [68], Abed and Esmaeel 2018 [49]
	Illumination variation using histogram equalization	Dange and Sayyad 2015 [69], Khirade and Patil 2015 [70], Malika and Vasanthi 2017 [71]
	Augmentation	Sladojevic et al. 2016 [72], Goncharov et al. 2019 [73]

TABLE 5. Details of segmentation technique used by various researchers.

Techniques Used		Author's
Edge-based segmentation	Canny edge detection, Sobel Operator, Prewitt Operator	Pydipati et al. 2006 [45], Anthonys and Wickramarachchi 2009 [75], Bankar et al. 2014 [76], Shinde et al. 2015 [77]
Thresholding Techniques	Otsu Thresholding, Adaptative Thresholding, Entropy Thresholding	Khirade and Patil 2015 [70], Pujari et al. 2016 [48], Cruz et al. 2018 [60], Das et al. 2020 [57]
Region Growing	Local Threshold	Pang et al. 2011 [78], Singh et al. 2015a [79]
Clustering	k-means	Rastogi et al. 2015 [80], Jadhav and Patil 2016 [81], Zhang et al. 2017b [82], Kaur et al. 2018b [56], Bashir et al. 2019 [35] Harakamanavar et al. 2022 [83] Zhang et al. 2018 [84]
	Fuzzy c-means	Jagtap and Hambarde 2014 [85], Bai et al. 2017 [86] Sahu & Pandey 2023 [87]
Grab Cut	-	Pantazi et al. 2019 [39]
Genetic Algorithm	-	Singh et al. 2015b [88], Singh and Misra 2017 [89]

4) FEATURE EXTRACTION

To discriminate one part of an image from another part, feature attribute extraction is one of the most important techniques in computer vision/ machine learning systems before classification. The features are advantageous to analyze objects and to allow the class label to an object. Generally, in the literature color, shape, and texture features are utilized for recognition. The classification of plant disease detection systems is enormously dependent upon the techniques used for image characteristic extraction [90]. From the literature, the spotted area segmentation and texture, color & shape feature extraction are of huge importance. Several plant disease detection researchers have used distinct convenient feature extraction techniques depending on texture, color & shape. Therefore, determining the correct features is a major issue in the detection of disease due to similarity in disease characteristics.

In computer vision/ machine learning applications, the selection of relevant features is also important to prevent overfitting and huge computational costs. Therefore, in computer vision and machine learning applications feature selection approaches are selected to find better relevant features from the extracted features. Some of the techniques such as PCA, genetic algorithm, and particle swarm optimization are used for feature selection. The literature utilizes various feature extraction techniques indicated in Table 6. By examining these approaches, an attempt is made to find the better-suited technique.

TABLE 6. Details of feature extraction technique used by various researchers.

Techniques Used		Author's
Feature Descriptor	GLCM, Wavelet Transform, Haralick feature, Gabor Transform, Local Binary Patterns	Mokhtar et al. 2015 [91],
	SURF	Bhagat & Kumar 2023 [92]
Texture Feature	GLCM Features	Bashish et al. 2011 [93], Tian et al. 2014 [94], Mainkar et al. 2015 [95], Islam et al. 2017 [96] Abed and Esmaeel 2018 [49], Sharif et al. 2018 [97], Deshapande et al. 2019 [47],
	Scale-invariant feature transform	Dandawate and Kokare 2015 [98], Mohan et al. 2016 [99], Ramesh et al. 2018 [100]
	Local Binary Pattern	Waghmare et al. 2016 [101], Singh et al. 2019 [102]
	Gabor Filter Transform	Gulhane & Gurjar 2011 [103] Prasad et al. 2012 [104] Kulkarni & R.K. 2012 [105] Jolly & Raman 2016 [106] Kaur et al. 2018b [56] Rao & Kulkarni 2020 [67]
	Wavelet Transform	Sabrol & Kumar 2016a [107] Gawali et al. 2017 [108]
	Histogram of Oriented Gradient	Dalal et al. 2005 [109] Bai et al. 2009 [110] Sannakki et al. 2013 [111] Pires et al. 2016 [112] Ramesh et al. 2018 [100] Kusumo et al. 2018 [113] Zamani et al. 2022 [114]
Color Feature	Color co-occurrence matrix	Pydipati et al. 2006 [45] Kai et al. 2011 [62] Revathi and Hemalatha 2014b [115] Ramakrishnan & Sahaya 2015 [116] Chouhan et al. 2019 [90]
	Color histogram	Caglayan et al. 2013 [117] Ramesh et al. 2018 [118]
Shape Feature	-	Anthonys & Wickramarachchi 2009 [75] Camargo & Smith 2009b [119] Wang et al. 2012 [120] Phadikar et al.2013 [121] Joshi & Jadhav 2017 [122] Sengar et al. 2018 [123] Sahu & Pandey 2023 [87]

The present scenario of literature during the last 12 years using different feature extraction techniques for distinct crops is represented in Fig 5. There are still various techniques to consider for plant disease detection using machine learning and computer vision techniques. The crop selection for such research is utilized by the availability of experts and data sets.

5) CLASSIFICATION

The utmost extrusive stage in computer vision/ machine learning is classification. The attainment of this step builds upon antecedent steps such as acquisition, preprocessing, and feature extraction/ selection. The manuscript is focused on disease detection in plants, the above steps are used for detection based on category and symptoms. In this system, a dataset is trained first and then used to classify the test image as healthy/defective. ''Machine learning (ML) is an application of artificial intelligence (AI) which provides you a system an ability to automatically learn, improve from experiences without being explicitly programmed and which can make decisions [124].'' ML techniques have been broadly distinguished as Supervised ML and Unsupervised ML. During the training process the labeled and unlabeled data work for supervised and unsupervised techniques respectively. The semi-supervised ML is another special supervised learning that works on both labeled and unlabeled data during the training process. In the literature, various classification techniques are utilized as indicated in Fig. 6.

Generally, support vector machine (SVM), artificial neural network (ANN), and k-nearest neighbor (k-NN) have been adapted for machine learning techniques [32]. Also, in a few studies vegetation index and fuzzy logic feature-based techniques are utilized. Distinct models of ANN like feed-forward NN, error backpropagation, multilayer perceptron, probabilistic neural network, and self-organizing map have been widely utilized for plant disease detection. Table 7 indicates the quality investigation of plant disease detection based on distinct classification techniques used by distinct researchers.

B. DEEP LEARNING USING IMAGE PROCESSING

Deep learning (DL) was first proposed in 1943 [146] as a hierarchy of machine learning for the detection of objects [147], [148], [149], [150], classification of images as well as processing of natural language [151], [152], [153]. DL forms high-level features by combining low-level features for distributed features discovering and sample data attributes. The ability of generalization and accuracy are enhanced compared to traditional ML techniques for target detection and image recognition. The development of DL is in three phases. Phase 1, during 1943-1969 was a linear model of neural network for classification. Phase 2, during 1986-1988 was a non-linear mapping of multi-layer perceptron for the BP (Backpropagation) algorithm. Phase 3, from 2006 to the present, has a ReLu activation function, ImageNet recognition, and deep learning model-AlexNet. The DL methods are

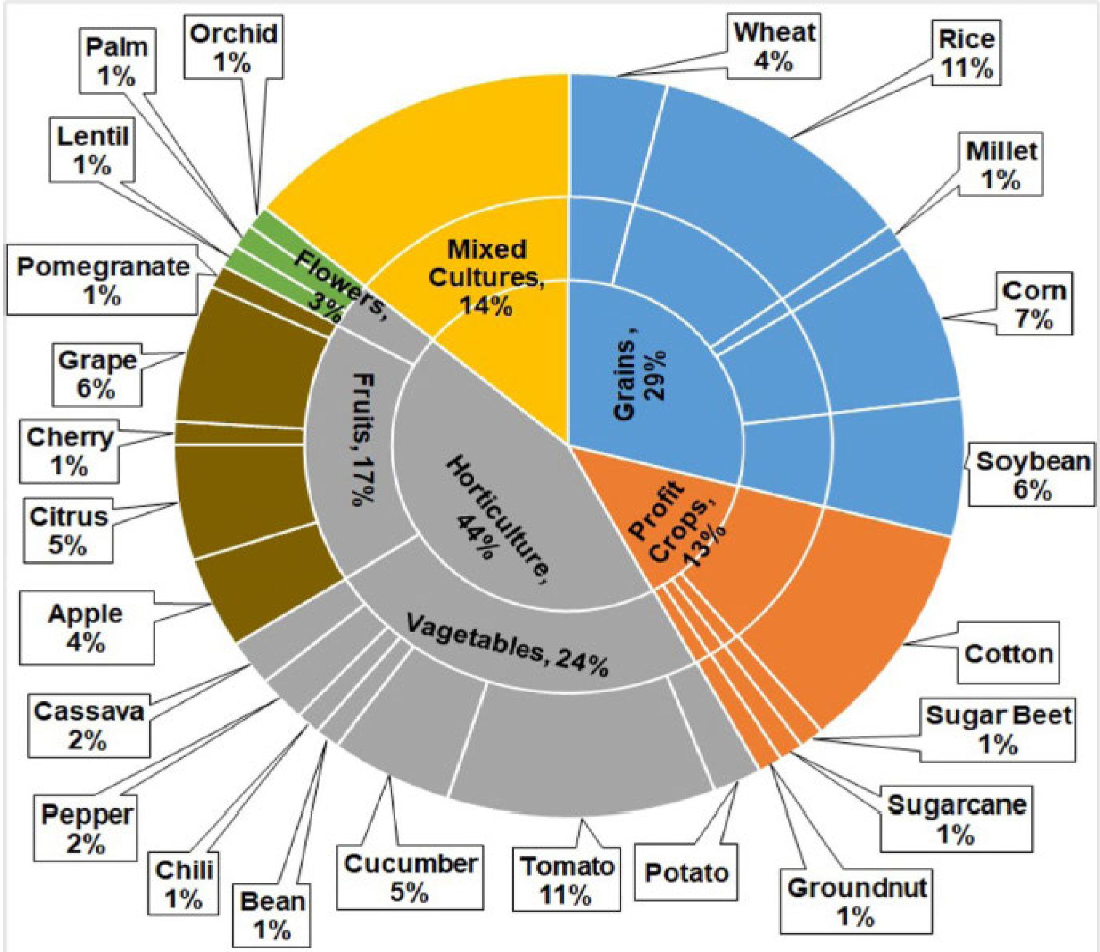

FIGURE 5. Different feature extraction techniques for distinct crops during last 12 years [16].

used for selecting features automatically. The techniques propose high-level features from the combination of low-level features. Subsequently, multifold advanced DL techniques proposed are shown in Fig. 7. Presently, the preeminent neural networks are convolutional neural networks, multilayer perceptron, and recurrent neural networks. The CNN establishes a classification model that incorporates AlexNet, GoogleNet, VGGNet, MobileNet, ResNet, and EfficientNet. AlexNet is the early model to adopt GPU devices for network training. The local normalization and rectified linear units are used as activation functions. Then VGG & GoogleNet use a few kernels with low memory [154]. In 2015, ResNet proposed by Microsoft uses connections and balance blocks [155]. In 2017, MobileNet was proposed by Google teams for embedded and mobile applications [156]. In 2019, EfficientNet was introduced by Google teams that use efficient coefficients to scale width, depth, and resolution. The deep network may not be used in plant disease detection since the basic models like VGG16 and AlexNet can fit the substantial accuracy needed. The C/C++, Python programming language can be used in the DL model. The various interface programming application such as deployment of the network, model support, and duplication of code is provided by the DL framework [157]. Currently, the widely used frameworks are Café, PyTorch, and Keras. The elemental steps of DL are indicated in Fig. 8 where CNN (Convolutional Neural Network) is used to incorporate features via convolutional, max-pooling, and fully connected layers. The features are extracted via the convolutional layer whereas texture and edges are extracted via shallow convolutional layer. The semantic information and complex texture features are extracted using the middle layer. The high-level semantic features are extracted using a deep layer. The max pool layer is used to hold the critical information in an image. Lastly, the fully connected layer is used to classify high-level features. Therefore, the implementation of DL requires the fundamental steps which are discussed in detail in the subsequent section.

1) IMAGE ACQUISITION

The data acquired must be accurate to obtain a perfect model. The DL database consists of training, validation, and test

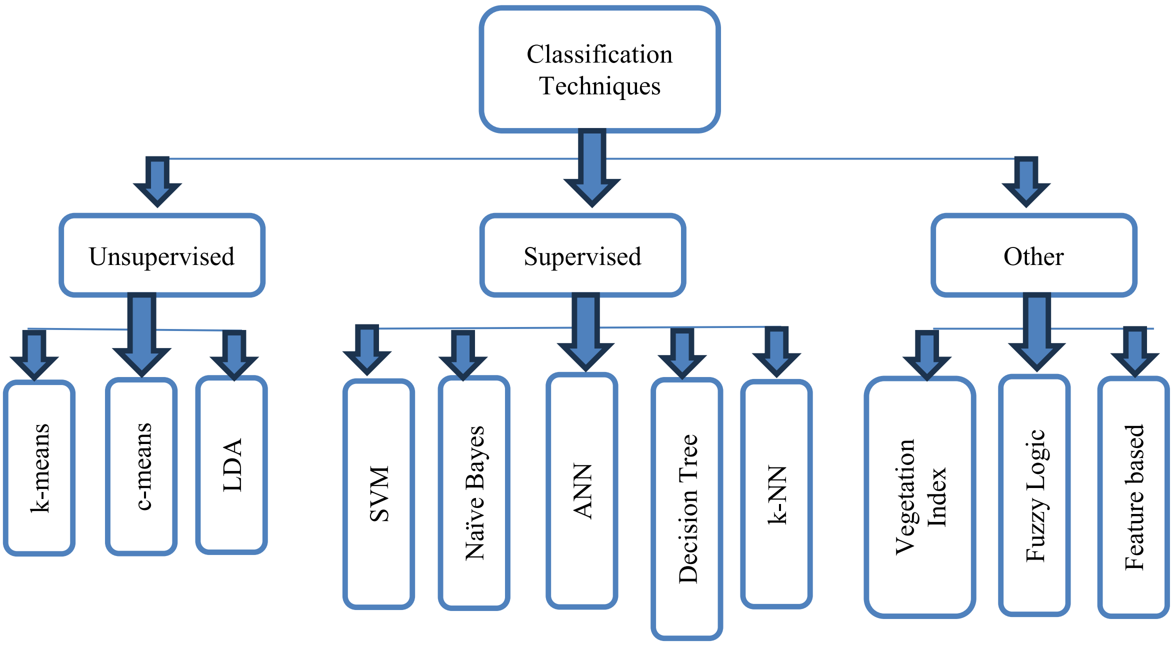

FIGURE 6. Distinct classifiers are analyzed in research for the detection of disease in plants.

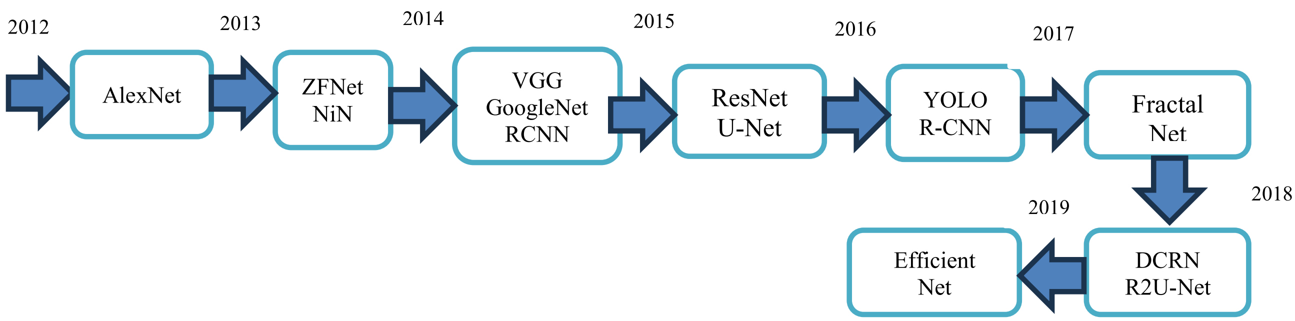

FIGURE 7. Hierarchy of DL techniques.

sets. The model is learned by the training set, hyperparameters are adjusted during validation and the performance evaluation is done by the test set [147], [158]. Various publicly available datasets are indicated in Table 8. These publicly available dataset is from two websites BIFROST (https://datasets.bifrost.ai/, accessed on 15 November 2023) and Kaggle (https://www.kaggle.com/datasets, accessed on 12 November 2023) which can be used for classification. In the literature of DL technique applied to plant disease classification, the most used public datasets are PlantVillage [33], [159] and Kaggle [160], many others also collect their own datasets [161], [162].

2) IMAGE AUGMENTATION

The plant leaf disease labeling, collection, and detection technique requires a huge number of datasets with financial and manpower resources. It is burdensome to collect short-onset period plants. The recognition rate will be destitute for small datasets in deep learning. Therefore, the expansion of the dataset is essential for plant leaf disease detection with deep learning. The practical application and the basic requirement can be removed using data augmentation. For example, the fundamental manifestation of plant disease is color. The original image color will not change by image enhancement. The two ways for data augmentation are conventional augmentation and generating an adversarial network.

Conventional techniques can boost the dataset via rotation, saturation adjustment, or symmetry of the mirror. Some of the advisory use augmentation techniques are AugMix [163], Fast AutoAugment [162], CutMix [164], population-based augmentation [165] and Rand Augment [166]. The flaws of this technique are deficient diversity, unevenness, and meager quality. The data augmentation technique is used by various researchers to improve the efficiency of the system as mentioned in Table 9.

TABLE 7. Details of classification technique used by various researchers.

	Authors	Dataset	Feature Extraction	Classifier	Accuracy
Rice	Joshi & Jadhav 2017 [122]	Agriculture Research (115)	Color & Shape	MD & k-NN	88.15%
	Mohan et al. 2016 [99]	60	Haar & SIFT	AdaBoost, k-NN & SVM	93.33%
	Zhang et al. 2018a [53]	-	Color	Vegetation Index	63.00%
	Bashir et al. 2019 [35]	APS (440)	SIFT	SVM	94.16%
	Shrivastava & Pradhan 2021 [125]	Real Field Images	-	SVM	94.65%
	Rath & Meher 2019 [126]	Real Field Images	-	Radial Basis NN	95.00%
Wheat & Corn	Luo et al. 2015 [127]	Chinese Academy (744)	Histogram	Histogram	94.44%
	Azadbakht et al. 2019 [50]	Hyperspectral data	Index based	Regression	95.00%
	Kusumo et al. 2019 [113]	Plant Village (3823)	SIFT, SURF	SVM, DT, RF, Naïve Bayes	87.00%
	Deshapande et al. 2019 [47]	Agriculture University Dharwad (200)	First Order histogram & GLCM	k-NN, SVM	88.00%
Soyabean	Gharge & Singh 2016 [128]	IPM Database (300)	GLCM	EBPNN	93.30%
	Pires et al. 2016 [112]	Federal (1200)	SIFT, SURF, HOG	SVM	96.25%
	Kaur et al. 2018b [56]	Plant Village (4775)	Color, Texture, Shape	SVM	84.00%
Millet	Caulibaly et al. 2019 [129]	Self (124)	Transfer Learning	VGG16	89.00%
Cotton	Rothe & Kshirsagar 2015 [51]	Self A460 Camera	Hus moments	EBPNN	85.52%
	Sivasangari & Indira 2015 [130]	Self	Color, Shape	SVM	99.30%
Sugar beet	Hallau et al. 2017 [131]	Self (1400)	Texture	SVM	82.00%
Groundnut	Ramkrishnan & Sahaya 2015 [116]	Self	CCM features	EBPNN	97.41%
Cane	Pujari et al. 2016 [48]	Self (9912)	RGB Color	SVM & EBPNN	92.00%
Mix	Sladojevic et al. 2016 [72]	Internet (33469)	-	CNN	95.80%
	Ferentinos 2018 [25]	Plant Village & Self (87848)	Transfer Learning	Alexnet, VGG	99.53%
	Arnal Barbedo 2019 [34]	Self (1575)	Transfer Learning	GoogLeNet	100.00%
	Pantazi et al. 2019 [39]	-	LBP	SVM	95.00%
	Rao & Kulkarni 2020 [67]	Plant Village	Gabor, Curvelet	Fuzzy Logic	90.00%
	Ahmed & Yadav 2022 [132]	Self-Created	GLCM	Random Forest	-

TABLE 7. ***(Continued.)*** **Details of classification technique used by various researchers.**

	Sahu & Pandey 2023 [87]	Plant Vilaage	Fuzzy c means	HRF SVM	-
Apple	Samajpati & Degadwala 2016 [133]	Self (80)	Color, Histogram, LBP, Gabor	Random forest	95.00%
	Jolly & Raman 2016 [106]	Self (320)	Haarlick & LBP	SVM	96.00%
Citrus	Sharif et al. 2018 [97]	Image gallery dataset (1000)	Color, Texture, Geometrical	Multiclass SVM	95.80%
Cheery	Sengar et al. 2018 [123]	Plant Village	Lesion Area	-	99.00%
Grape	Waghmare et al. 2016 [101]	Self (450)	HSV, LBP	Multiclass SVM	89.30%
	Cruz et al. 2018 [60]	Self (272)	-	AlexNet, Inception-v3	98.00%
	Goncharov et al. 2019 [73]	Plant Village (2986)	-	Siamese	92.00%
	Javidam et al. 2023 [134]	Self	GLCM	SVM	98.97%
	Shantkumari et al. 2023 [135]	Plant Village	-	Improved k-NN	-
Pomegranate	Khot et al. 2016 [58]	-	HSI	Minimum distance classifier	-
Palm Oil	Masazhar & Kamal 2018 [46]	-	GLCM	Multiclass SVM	95.00%
Lentil	Singh et al 2019 [102]	300	LBP	Visual Examination	-
Potato	Islam et al. 2017 [96]	Plant Village (300)	Color, Texture	Multiclass SVM	95.00%
	Patil et al. 2017 [136]	Plant Village (8920)	Texture	SVM, ANN, RF	92.00%
	Verma et al 2020 [137]	Plant Village	-	Capsule Network	91.83%
Bean	Abed & Esmaeel 2018 [49]	100	GLCM	SVM	100.00%
Cucumber	Zhang et al. 2017b [82]	Self (300)	PHOG	SVM	91.48%
	Krishna Kumar & Narayan 2019 [138]	Real Field Images	-	k-means SVM	86.00%
Tomato	Raza et al. 2015 [139]	Self (71)	Pixel Value	SVM	90.00%
	Sabrol & Kumar 2016a [107]	Self (180)	Color moments, Histogram	ANFIS, FF-BPNN	87.20%
	Fuentes et al. 2017 [40]	Self (5000)	DWT, Haar Wavelet	ResNet-50, VGG16	83.06%
	Ashqar & AbuNaser 2018 [140]	Plant Village (9000)	Transfer Learning	CNN	99.84%
	Karadag et al. (2018) [37]	ARI (80)	Wavelet	ANN, NB, k-NN	84.00%
	Das et al. 2020 [57]	Self-Created	Mean, Entropy	SVM, k-NN	87.60%

TABLE 7. *(Continued.)* **Details of classification technique used by various researchers.**

	Panigrahi et al. 2020 [141]	Self-Created	-	RF	79.23%
	Sujatha et al. 2021 [142]	Self-Created	-	SVM	87.00%
	Zamain et al. 2022 [114]	Plant Village	PCA	SVM	-
	Harakamanavar et al. 2022 [83]	-	k- means	SVM	88.00%
	Rahman et al. 2023 [143]	Self-Created	GLCM	SVM	88.00%
	Bhatia et al. 2020 [144]	Mildew dataset	-	Extreme Learning	89.19%
	Bhatia et al. 2021 [145]	Mildew dataset	-	SVM Logistic Regression	92.73%

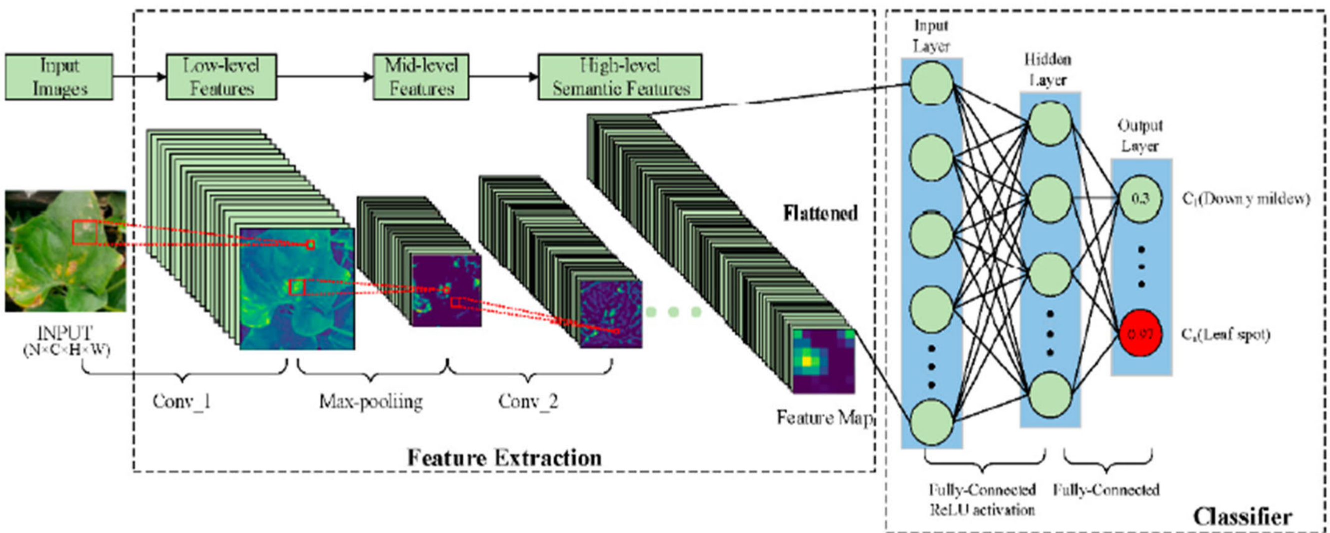

FIGURE 8. **Elemental steps of deep learning.**

TABLE 8. **Bulky publicly available dataset used in deep learning.**

Dataset Name	No. of images	Classes	Source
New Plant Disease Dataset	87000	38	Kaggle
Plant Village Dataset	162916	38	Kaggle
Flowers Recognition	4242	4	Kaggle
Plant Seedings Dataset	5539	12	BIFROST
Weed Detection in Soybean Crops	15336	4	Kaggle
Plant Pathology Challenge	3651	4	Kaggle

TABLE 9. **Various literature regarding data augmentation.**

Authors	Dataset	Techniques	Accuracy increases by
Bin et al. 2017 [168]	1053 to 13689	PCA	4.00%
Lin et al. 2019 [162]	10820 to 32460	Radial blur	3.15%
Ferentinos 2018 [25]	54309 to 87848	Cropping, Resizing	5.23%
Arnal Barbedo 2019 [34]	1567 to 46409	Segmentation	12.00%
Fuente et al. 2017 [48]	5000 to 43398	Resizing, Cropping	3.87%
Srdjan et al. 2016 [169]	4483 to 33469	Rotation	3.00%

Goodfellow et al. 2014 [167] propose a generating model known as Generate Adversarial Network (GAN). Consequently, huge variations of GAN have developed separately like CGAN, DCGAN, LAPGAN, PGGAN, InfoGAN, F-GAN, WGAN, LeakGAN, SeqGAN. The objective is to generate samples of a dataset with the equivalent attributes. This model primarily contains a generator and discriminator as shown in Fig. 9. The GAN technique is used by various researchers to improve the efficiency of the system as mentioned in Table 10.

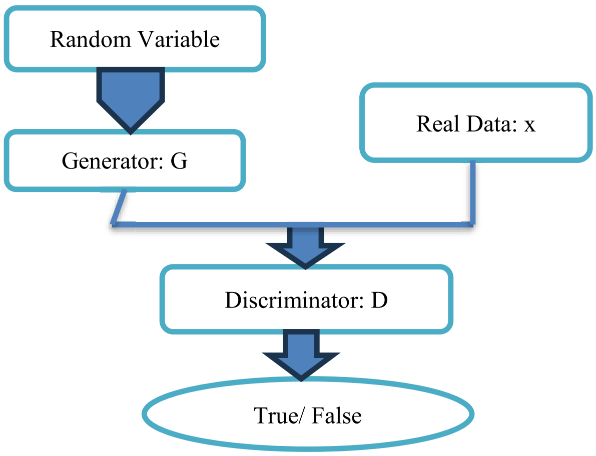

FIGURE 9. Generate adversarial network architecture.

TABLE 10. Various literature regarding data augmentation.

Authors	Dataset	Techniques	Accuracy increases by
Nazki et al. 2020 [170]	2789	ARGAN	5.20%
Tian et al. 2019 [171]	TeaImage	CycleGAN	28.00%
Wu et al. 2020 [172]	GoogleNet	DCGAN	94.33%
Liu et al. 2020 [173]	Grape	LeafGAN	-

3) IMAGE CLASSIFICATION

Recently, several fruitful applications of DL for the classification of disease in plants have been established. Nonetheless, transparency and interpretability are curtailments in classification using DL. The mechanism consists of a black box beyond one minutiae or explanation. The disease detection system must reveal the disease, and symptoms as well as achieve high accuracy. Various researchers utilize distinct techniques using DL as indicated in Table 11.

C. HYPERSPECTRAL IMAGING USING IMAGE PROCESSING

The disease in plants is burdensome sometimes via computer vision systems or naked eyes due to pathogens' growing process [221]. The hyperspectral imaging sensor is primarily fixed in the electromagnetic spectrum range of infrared and visible (400 nm ∼ 2500 nm). These sensors attain shadowy information from a wide range of bands [222]. This range is eminently sensitive to leaf variation caused by distinct leaf disease such that antecedent pathology detection can be done. Consequently, hyperspectral imaging targeted plant disease detection at the early stage. Wang et al. 2019 [149] propose a GAN for hyperspectral imaging as OR-AC-GAN (Outlier Removal—Auxiliary Classifier-GAN) for tomato leaf disease detection and achieve 96.25% accuracy as shown in

TABLE 11. Details of DL technique used by various researchers.

Authors	Dataset	DL Technique	Accuracy
Mohanty et al. 2016 [33]	54306 images, 26 diseases	AlexNet, GoogLeNet	99.35%
Fuentes et al. 2017 [40]	5000 images,	VGGNet, ResNet	83.00%
Bin et al. 2017 [168]	13689 images, 4 diseases	AlexNet	97.62%
Brahimi et al. 2017 [159]	14828 images, 9 diseases	GoogLeNet	99.18%
Ale et al. 2019 [174]	87848 images	VGG	99.53%
Darwaish et al. 2019 [175]	15408 images, 3 diseases	VGG16 & VGG19	98.20%
Yuwana et al. 2019 [176]	5632 images,	MCT	-
Chen et al. 2020 [177]	900 images, 2 categories	VGGNet, Inception	92.00%
Mishra et al. 2020 [178]	-	DCNN	88.46%
Ashok et al. 2020 [179]	Self-Created	AlexNet	95.75%
Trivedi et al. 2020 [180]	Plant Village	CNN	95.81%
Jasmin & Tuwaijari 2020 [181]	Plant Village	CNN	98.29%
Panchal et al. 2021 [182]	Plant Village	CNN	-
Chug et al. 2023 [183]	IARI Dataset	Hybrid DL	100% 99.64%
Wang et al. 2019[184]	Internet Image	RCNN with VGG16	98.00%
Picon et al. [185]	Real Field	ResNet50	86.00%
GadeKallu et al. 2021 [186]	Plant Village	DNN	97.00%
Nitish et al. 2020 [187]	Plant Village	ResNet50	93.40%
Verma et al. 2020 [188]	Plant Village	AlexNet	94.00%
Wspanialy & Moussan 2020 [189]	Plant Village	ResNet, U-Net	98.00%
Karthik et al. 2020 [190]	Plant Village	CNN	99.60%
Ahmad et al. 2020 [191]	Laboratory & Field Dataset	ResNet, InceptionV3	98.00%
Lee et al. 2020 [192]	Plant Village	GoogleNet, VGG16	98.40%
Atila et al. 2021 [193]	Plant Village	EfficientNet	91.20%

TABLE 11. ***(Continued.)*** **Details of DL technique used by various researchers.**

Agarwal et al. 2021 [194]	Plant Village	CNN	98.46%
Ngugi et al. 2020 [195]	Real Condition Dataset	CNN	93.75%
Chen et al. 2021 [196]	Plant Village	Enhanced ANN & CNN	93.45%
Krishnan et al. 2022 [197]		CNN with Fuzzy	99.16%
Singh et al. 2022 [198]	Plant Village	CNN AlexNet	98.83%
Elaraby et al. 2022 [199]		CNN with PSO	
Ashwin Kumar et al. 2022 [200]	Plant Village	MobileNet	98.70%
Pandian et al. 2022 [201]	Self-Created	DCNN	98.41%
Mahum et al. 2022 [202]	Plant Village	DenseNet201	97.20%
Haridson et al. 2022 [203]	-	ReLu Softmax	91.45%
Algani et al. 2022 [204]	Plant Village	CNN	-
Huang et al. 2022 [205]	Self-Created	Fully Convolutional	97.59%
Yu et al. 2022 [206]	Plant Village	Inception Convolutional	98.17%
Khan et al. 2022 [207]	Self- Created	Entropy ELM	98.40%
Yu et al. 2022 [208]	Apple	MSO ResNet50	95.70%
Khan et al. 2022 [209]	Self-Created	DL	88.00%
Algani et al. 2023 [210]	-	ACOCNN	99.98%
Shoaib et al. 2023 [211]	2300 images	ResNet	99.62%
Singh et al. 2023 [212]	Self-Created	CNN	99.39%
Sharma et al. 2023 [213]	Self-Created	DLMN Net	96.56%
Daniya & Vigneshwari 2023 [214]	Self-Created	RHGSO	93.04%
Arun et al. 2023 [215]	Plant Village	Concatenated DL	98.14%
Pal & Kumar 2023 [216]	Plant Village	Inception Visual	-
Reddy et al. 2023 [217]	Plant Village	ResNet50	99.73%
Kaya et al. 2023 [218]	Plant Village	Dense Net	98.17%
Thai et al. 2023 [219]	Cassas leaf	Pruning Network	-
Thakur et al. 2022 [220]	Plant Village	CNN	99.16%

Fig. 10. Various researchers use distinct techniques for HSI plant disease detection are illustrated in Table 12.

TABLE 12. **Details of hyperspectral image classification used by various researchers.**

Authors	Disease	Algorithm	Accuracy
Moshou et al. 2005 [223]	Rust	QDA	94.50%
Mahlein et al. 2012 [224]	Mildew powdery	SAM	98.90%
Huang et al. 2012 [225]	Rust	Linear regression	-
Nagasubramanian et al. 2017 [226]	Rot	3D CNN	95.73%
Alisaac et al. 2018 [227]	Head blight	SVM	99.00%
Abdulridha et al. 2019 [228]	Canker	RBF	96.00%
Ngyugen et al. 2020 [229]	-	MLP	99.00%
Feng et al. 2021 [230]	Rot	Deep domain	88.00%
Wan et al. 2022 [231]	Pathogen infected	-	-
Bhujade & Sambhe 2022 [232]	-	DL	-
Cui et al. 2022 [233]	Valsa Canker	DC-CNN	98.00%
Wu et al. 2022 [234]	Plant Village	Adaptive Coherence	97.41%
Fajardo et al. 2022 [235]	Banana Leaf	Logistic Regression	-
Yong et al. 2023 [236]	Basal Rot	RCNN	91.93%
Wu et al. 2023 [237]	Rust	ELM	96.67%

An extensive number of modern developments in different pathogen systems using distinct group of extremely sensitive sensors and various data analysis pipeline have been summarized in Fig 11 and Table 13. It overviews the current sensors technologies used for automatic identification and detection of plants. These sensors can be implemented in agriculture applications of plants. Depending on the scale, distinct platforms of plants can be managed and observed.

RGB Imaging: RGB (Red Green Blue) is the simple source og digital images for disease detection and identification. The technical parameters such as photo sensor for light sensitivity, optical focus and spatial resolution has been improved significantly [238].

Multi and Hyperspectral Sensors: Multispectral sensors assess the spectral information of object in various wavebands [239]. It also provide RGB and near infrared wavebands [240]. Hyperspectral sensors provides spatial and spectral information from the object. The spatial resolution depends upon the object and the sensor [241].

Thermal Sensors: Infrared thermography (IRT) assess plant temperature and is correlated with the status of plant water, the microclimate in crop stand [242]. IRT can be used in plant science at distinct spatial and temporal scales from small scale applications [238].

Fluorescence Sensors: Chlorophyll fluorescence sensors are active sensors with laser light or LED source that assess photo synethitic transfer [243]. It combine with image analysis technique have been useful for quantification and discrimination of fungal diseases.

TABLE 13. Details of OPTICAL SENSORS used by various researchers.

Authors	Sensors	Crop	Disease
Neumann et al . 2014 [244]	RGB	Sugar beet	Angular Bacateria
Bergstrasser et al. 2015 [245]	Spectral Sensors	Sugar beet	Leaf Spot
Wahabzada et al. 2015 [246]	Spectral Sensors	Barley	Brown Rust
Polder et al. 2014 [247]	Spectral Sensors	Tulip	Virus
Berdugi et al. 2014 [248]	Thermal Sensor	Cucumber	Mildew
Gomez 2014 [249]	Thermal Sensor	Rosa	Mildew
Bauriegel et al 2014 [250]	Fluorscence	Lettuce	Mildew
Rousseau et al. 2013 [251]	Fluorscence	Bean	Bacterial Blight

TABLE 14. Details of FEW SHOT image classification used by various researchers.

Authors	DL Frame	Dataset	Sample
Medela et al. 2020 [255]	One-shot learning	Medical	60
Ren et al. 2019 [256]	One-shot learning	PlantVillage	20
Huang et al. 2020 [257]	Few-shot learning	PlantVillage	1
Nagasubramanian et al. 2017 [258]	Few-shot learning	MiniImageNet	15
Alisaac et al. 2018 [259]	One-shot learning	Self-Acquired	1160
Abdulridha et al. 2019 [260]	CDCGAN	Self-Acquired	40
Li & Yang 2021 [261]	Few Shot	Self-Created	-
Li & Chao 2021 [262]	Few Shot	Plant Village	10
Lang 2021 [263]	Few Shot	Cotton leaf	-
Wang et al. 2022[264]	Few Shot	Self-Created	-
Whusquiza et al. 2022 [265]	Few Shot	Self-Created	-
Garg & Singh 2023 [266]	Few Shot	Plant Doc	30
Liu et al. 2023 [267]	Few Shot	Plant Village	-
Li et al. 2023 [268]	Few Shot	Plant Village	10

D. FEW SHOT LEARNING (FSL) USING IMAGE PROCESSING

A new model of machine learning that facilitates learning from machines using a limited number of labeled databases is known as FSL [252]. The experiment's purpose and problem complexity determine the number of shots required. FSL uses datasets with prior knowledge for training. This process includes data, models, and optimization to tune the samples. The training phase leads to creating source tasks activated by obtaining multiple tasks from single or multiple datasets as represented in Fig. 12. The machine learning model is combined with these source tasks to train the system. The most commonly available dataset for plant recognition systems consists of several thousands of images. The performance of the system depends on the quantity as well as the quality of images. Therefore, FSL is the best possible feasible solution for a confined number of samples [253], [254]. Various researchers use distinct techniques for small samples-based plant leaf disease detection are illustrated in Table 14.

Various applications for FSL to abate parameters are mentioned below.

i. **Embedding-** It features an extraction technique also known as data transformation benefitted for low dimension reduction (higher dimensional map to lower dimension). This includes pre-trained CNN with ImageNet for extraction of features (reduce training time) and SVM for classification. SVM uses distinct kernels to handle complex systems [268].

ii. **Multitask Learning-** The process includes a shared model to train multiple tasks generally classified as hard parameter sharing and soft parameter sharing. In HPS certain parameters are shared to the entire task and other parameters learn precise features. In SPS each parameter is shared with individual tasks. This multitask learning incorporates all the models that can be trained for all the tasks rather than a single task [269].

iii. **Transfer Learning**- The process was initiated to accomplish a diminished training period by reusing gathered knowledge [270].

iv. **Meta-Learning**- This refers to optimization and performance improvement via multiple-task experience. It is also known as support sets that learn from multiple sources instead of a single source (conventional machine) alongside FS to adequately develop a model for clarifying query sets.

The meta-learning and multitasking learning are generally unstable and marginally upgraded by a few proportions. Transfer learning is the most superior and stable to train the model from scratch. These approaches are computationally costly compared to the degree of performance for enhancement. The future work led to enhancing existing FSL algorithms to cooperate with plant disease characteristics recognition. Further comparative and comprehensive studies are done for the performance evaluation of each FSL approach.

E. MOLECULAR DIAGNOSIS TECHNIQUES

Pest and disease may heighten the total food production losses by 35%. Globally, under quarantined pathogens, some of the austere plant diseases are nematodes, parasitic, oomycetes, viruses, bacteria, and fungi.

FIGURE 10. Segmentation comparison (a) Direct CNN (b) AC-GAN (c) OR-AC-GAN.

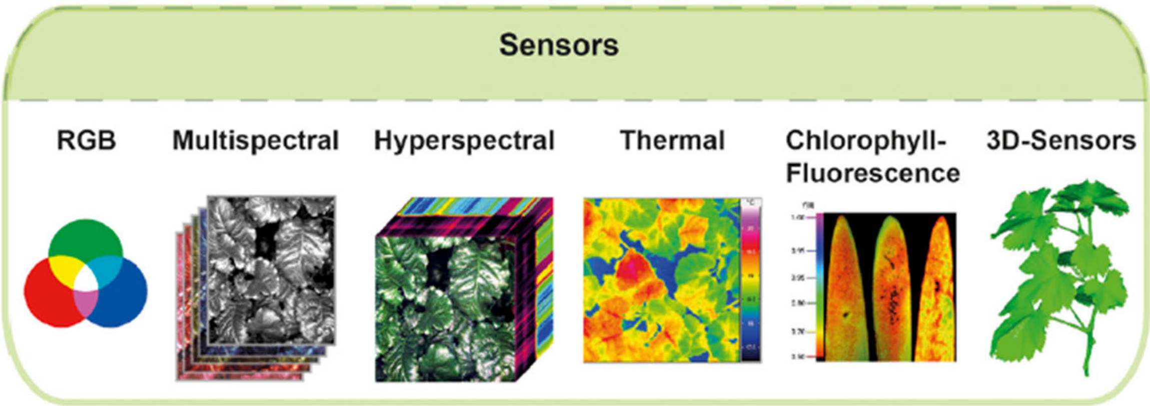

FIGURE 11. Overview of sensor technologies [238].

FIGURE 12. Approach to imitate multiple sources.

Fang and Ramasamy 2015 [15] reported early disease diagnosis is critical to overcome outbreaks of disease. Scientists utilize laboratories and certain advanced facilities compared with traditional techniques resulting in reliable diagnosis methods. Nonetheless, these approaches performed better only with severe infection in plants. Therefore, to bridge this gap certain adequate molecular diagnosis techniques like management control on decisions or tools for plant disease overcome the traditional techniques. The emerging molecular technologies to diagnose plants due to the necessity of managing diseases are as follows.

i. **Enzyme-linked immunosorbent assay (ELISA):** This approach is widely employed as a diagnosis device in plant and chemical pathology. It identifies and detects the particular substance in the sample, and provokes a color change that expands antibodies with enzymes [271]. ELISA is not very sensitive compared to other methods [272] but is labor-intensive and sensitive [273]. Despite antibodies like polyclonal being adequate to distinguish pathogens, nevertheless, these are not ample specific. One other antibiotic monoclonal is innumerable specific but further costly. As ELISA is tremendously sensitive to viruses results in the successful development of antibodies [274]. The technique will be inadequate to detect bacteria due to low sensitivity [15]. Therefore, pathogens bacteria, and fungi for rice disease are not effective using ELISA. Also due to inadequacy of specificity, ELISA is not able

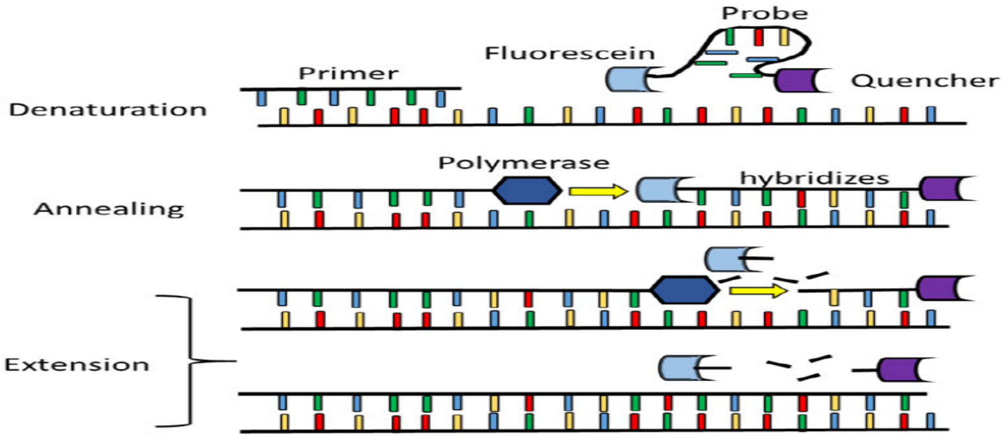

FIGURE 13. Illustrative real-time PCR schematic [286].

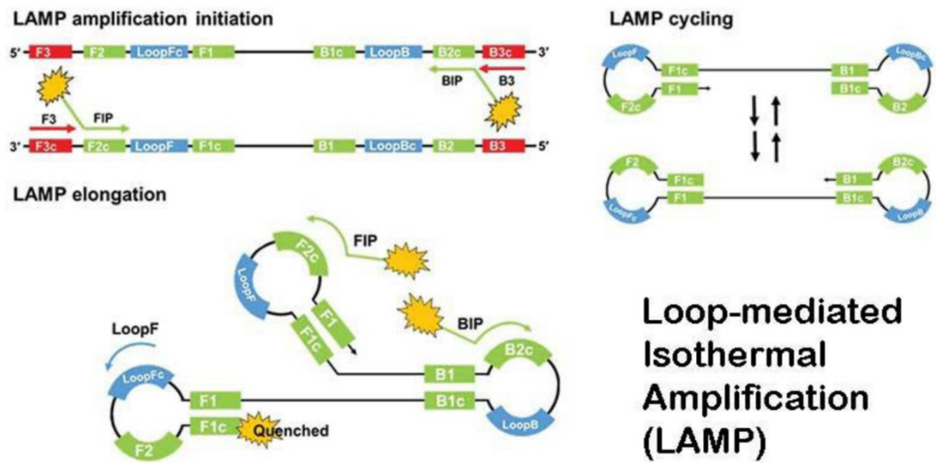

FIGURE 14. Amplification of LAMP [283].

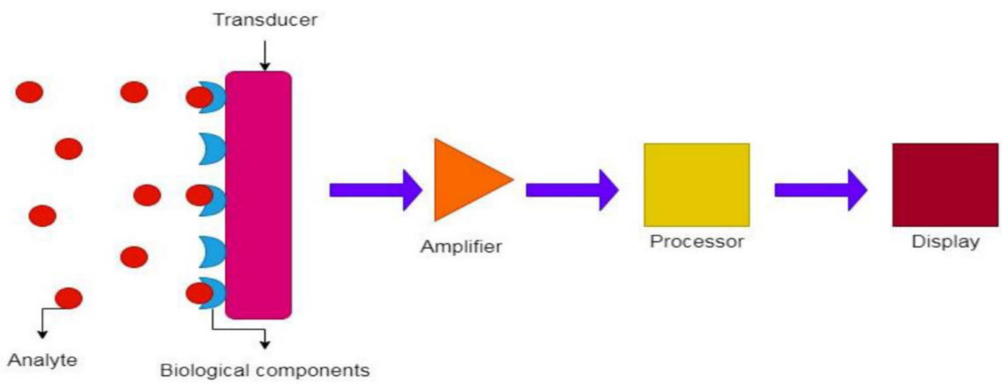

FIGURE 15. Biosensors & its process schematic illustration [286].

to discriminate several strains with luminous specific symptoms [275].

ii. **Conventional PCR**: The most valuable achievement in molecular biology is the Polymerase Chain Reaction (PCR) established in the mid-80s [276]. The technique requires various components and reagents which incorporate monovalent cation, primers, buffer solution, triphosphate deoxynucleotide, and polymers.

The technique anticipates electrophoresis used for the detection of plant pathogens. The nucleic acid evocation technique affects the PCR sensitivity and (Dioxyribo Nucleic Acid) DNA target variability [277]. The highly sensitive nature of PCR results in DNA amplification. The PCR limits contamination susceptibility and inadequacy of robustness for a long time with false positive results.

iii. **Real-Time PCR**: A significantly revolutionary technique developed by Mullis Karry in 80 is also known as a quantitative polymerase chain reaction. This technique diminishes the cross-contamination risk with most precipitate-sensitive detection [278]. The infected tissues and vectors can easily detect DNA pathogens using RT-PCR [279]. This technique points exclusively to known genes [280] with sequence data indicated in Fig. 13.

iv. **Lamp:** The most common alternative to RTPCR is a lamp due to its cost-effectiveness, simplicity, rapidity, and practicality. Under isothermal elaboration, lamps recognize HBV (Hepatitis B Virus) using certain specific sequences of elaborated DNA [281]. Lamp contains four distinct primers two for inner and outer each (Forward inner primer, backward inner primer, forward outer primer, backward outer primer) as indicated in Fig. 14. A loop is formed for high displacement and auto strand DNA for 60 minutes at 65 degrees C isothermal situation [282]. As indicated in the figure the process of corporate initial material production, amplification, recycling, and elongation [283]. Lamp detects amplicon using the detection of colorimetric, fluorimeter, and turbidimeter [284]. The technique incorporates the formation of primer to cultivate the 65-degree C temperature [285]. The technique is highly sensitive for elaboration therefore proper practice and handling are enforced to escape the contamination risk.

v. **Biosensor**: An ample range of challenges and problems in pharmacology medicine, security, food safety, and agriculture can be resolved using biosensors. Clark and Lyons [287] develop biosensors via measurement of glucose using hydrogen peroxide or oxygen electrochemical detection. Vigneshvar et al. 2016 further evolved biosensors into innovative approaches like nanotechnology, and electrochemistry.

"According to International Union of Pure and Applied Chemistry recommendations (1999), a biosensor is described as a device that deploys the integration of specific biochemical reactions mediated by isolated enzymes, immune systems, tissues, organelles or whole cells to detect chemical compounds, commonly by electrical, thermal or optical signals". These analytical appliances are constituted of the transducer coupled with a biorecognition component and disciple into the analytical signal as shown in Fig. 15. The advantage of the device includes cost effectiveness, sensitivity, and simplicity.

vi. **Next Generation Sequencing (NGS):** Generally, the sequencing mechanism depends on molecular detection approaches [288]. The first generation of commercial and laboratory sequencing was employed due to poor radioactivity and improved efficiency [289]. The approach is ineffective and tedious for large sequences and efficient for short sequences. The NGS is also known as a parallel massive sequence capture sequence in less time, low cost, and high throughput [290]. The unique biology molecular opportunities are incorporated via NGS such as mela genomics [291], single-cell sequencing, sew protein [292], and whole genome sequencing [293]. The approach limits to diagnosis of helm pathogen [294], [295]. Advanced NGS adopts colony sequencing, parallel massive sequencing, and pyrosequencing by legation nucleotide [287].

V. LIMITATIONS

While plant disease detection has been analyzed automatically, there are several limitations in each step or process. The samples are very difficult to obtain for particular diseases [37], [38], [39]. It is observed that system performance is immensely influenced by the type of dataset used i.e. laboratory controlled or real-time dataset. The data obtained in an uncontrolled environment results in increased system complexity but it is of huge significance in agriculture development and more adequate in today's research. Several difficulties have been observed during the extraction of features [106]; similarity in the infected area leads first to the extraction of inappropriate features and then false classification based on irrelevant feature matching. Thus, it is necessary to select a set of features because each feature has a different importance level. Various classification techniques used for plant disease detection are SVM, ANN, Naïve Bayes, backpropagation neural network, decision tree, and k-nearest neighbor [123], [124], [125], [126]. Despite these classifiers convolutional neural networks have been explored for huge samples (thousands and more) more accurately. However, the overfitting of CNN is seen as a major issue in systems about plant disease detection using deep neural networks [195]. Various approaches have been observed which can improve the efficiency of convolutional neural network-based classification systems for multiple cultures.

According to the literature, the proposed system has a set of specifications to be fulfilled necessarily, if any specification is unconsidered then the system proposed provides inaccurate disease detection. Therefore, a versatile system must be designed by a researcher with an adjustable set of specifications instead of a constant. The actual use of machine learning has been disturbed due to overfitting. Hence a highly generalized and adaptable system is to be developed for plant disease detection in various cultures. Machine learning is the most popular domain due to the availability of techniques and tools, but these resources must not be compromised for accuracy.

VI. CHALLENGES

From the literature review, it can be found that one of the challenges facing plant disease detection is the lack of experts to annotate accurately. The issue arises when experts cannot differentiate correctly between dead plants and disease-infected plants. This process requires experienced and expert professionals to detect diseases in plants that are costly and difficult, especially for rare or new diseases. Furthermore, using DL techniques for modeling, tuning, and resource training is another important challenge. The shallow architectures are best suited for small datasets. The most recent models for disease detection offer another angle to consider in building a model for plant disease detection. The ML, DL, and FSL are to be considered to enhance the detection models. This paper recommends that future research on plant disease detection, classification, and quantification of disease will improve mart agriculture.

Various issues and factors that may affect plant disease classification and identification are listed below:

(i) The plant disease detection system performance mainly depends on the feature extraction and classification technique used [144].

(ii) Most of the literature reviewed utilizes Plant Village Dataset i.e. laboratory images rather than real time images. The classifier performance heavily depends on the dataset used for testing and training [144].

(iii) The real field images may have complex backgrounds; therefore, segmentation of affected areas is difficult which affects the performance of the system [296].

(iv) The plant has a deficiency in nutrients [74] and contamination at an early stage of development [8].

(v) The infected area estimation and severity management with disease detection may be used to control pesticides [188].

(vi) The real-time illness efficiency on constrained devices could be considered [189].

(vii) Various diseases may create distinct manifestations simultaneously. Therefore, it will be difficult to identify and combine hybrid symptoms.

(viii) The hyperparameters tuning and selection have the potential to have a powerful influence on performance [196].

(ix) Due to the characteristic of disease uniformity and attribute selection, disease identification systems face significant challenges [297].

VII. CONCLUSION

The manifestation of imminent pathogens plants extends to become a considerable threat to food safety, the ecosystem, and the global economy. Moreover, crucial factors like mobility globalization, vectors, changes in climate, and evolution in pathogens emboldened the advancement of invasive pathogen plants. An adequate automated approach is eminently desired to bury losses in agriculture. Therefore, an automated plant disease recognition and classification is reviewed using machine learning, deep learning, and few-shot learning. The review presents various well-known approaches such as acquisition, preprocessing, segmentation, feature extraction, and classification. Despite RGB images some of the studies use hyperspectral imaging for plant leaves. The hyperspectral does not require a labeled database and microscopic symptoms are easily detected. These automated techniques contribute a manifestation of research in accessible time. Also, the diagnosis molecular tools with the latest techniques for disease detection are reviewed. The entire techniques indicated in this paper are gratuitous to the sensitive, specific, and rapid detection of plants. Moreover, in the future to detect disease in plants, the combination of severe side program and client mobile as well as electrophysiology would be exceptional future research guidance.

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